

Modelling fundamentals: a linear model with a single, continuous

BIOL2022 - L02b

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Learning objectives

By the end of this lecture, you should be able to:

- Describe a linear model for a continuous response and predictor.
- Write and interpret a simple linear model equation.
- Explain how residuals and the least-squares method are used to fit a line.
- Interpret coefficients, confidence intervals, and R^2 in model output.
- Recognise that coefficient interpretation depends on residual checks, which we will learn next week.

Introduction to linear modelling

“Cars with flames painted on the hood might get more speeding tickets. Are the flames making the car go fast? No. Certain things just go together. And when they do, they are correlated. **It is the darling of all human errors to assume, without proper testing, that one is the cause of the other.**”

— Barbara Kingsolver, *Flight Behavior* (2012)

A short history of least squares



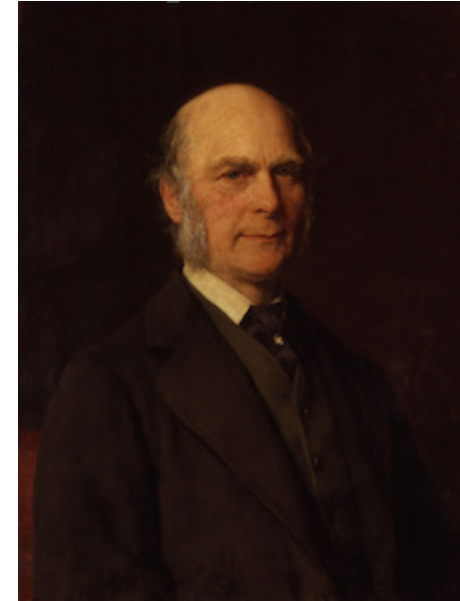
Illustrative image of Adrien-Marie Legendre

Legendre: No known portrait exists; this is an illustrative image from [Wikipedia](#). Introduced least squares in 1806.



Carl Friedrich Gauss

Gauss: Published work on least squares in 1809. [Source: Wikipedia](#)



Francis Galton

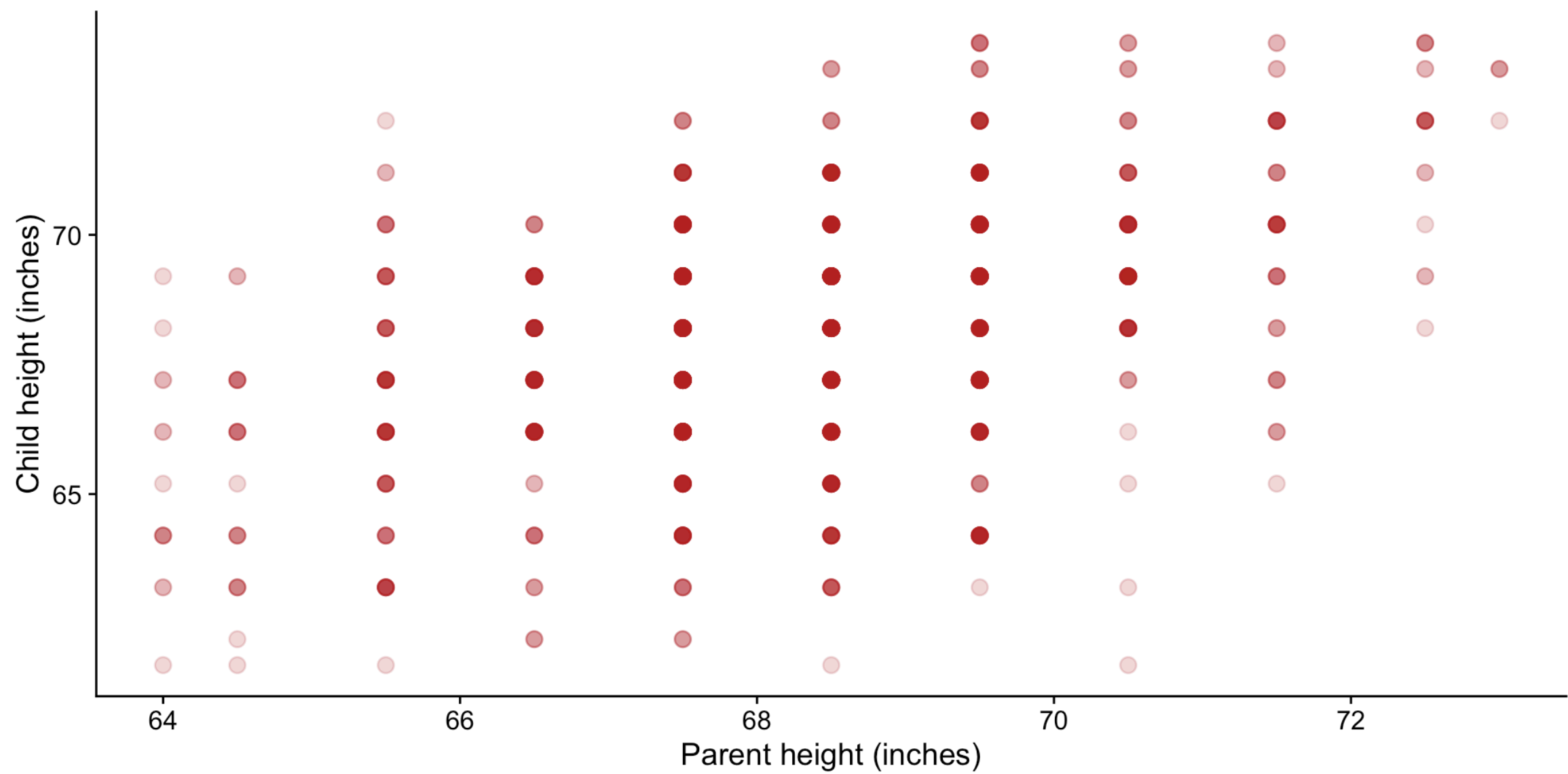
Galton: Introduced regression to the mean ([1886](#)) and correlation ([1888](#)).

The Galton dataset

- 928 children from 205 parent pairs.
- Parent and child heights were recorded in inches.
- Heights were recorded in size classes, so the plot will show horizontal bands.

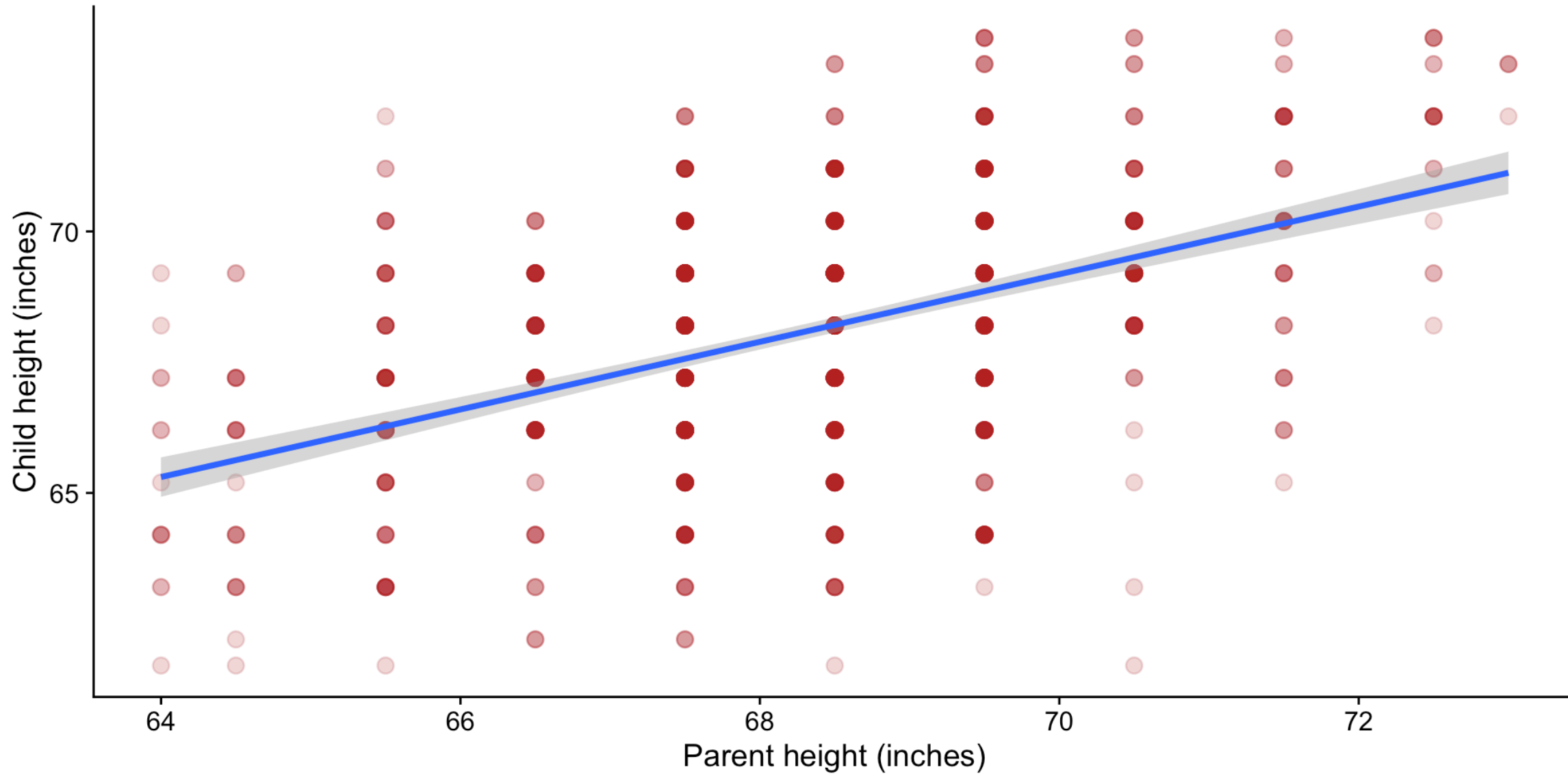
Question: How does child height vary with parent height?

Scatterplot



We want to explain this *noisy* relationship...

Regression line



We want to explain this *noisy* relationship... using a linear model.

How does linear modelling work?

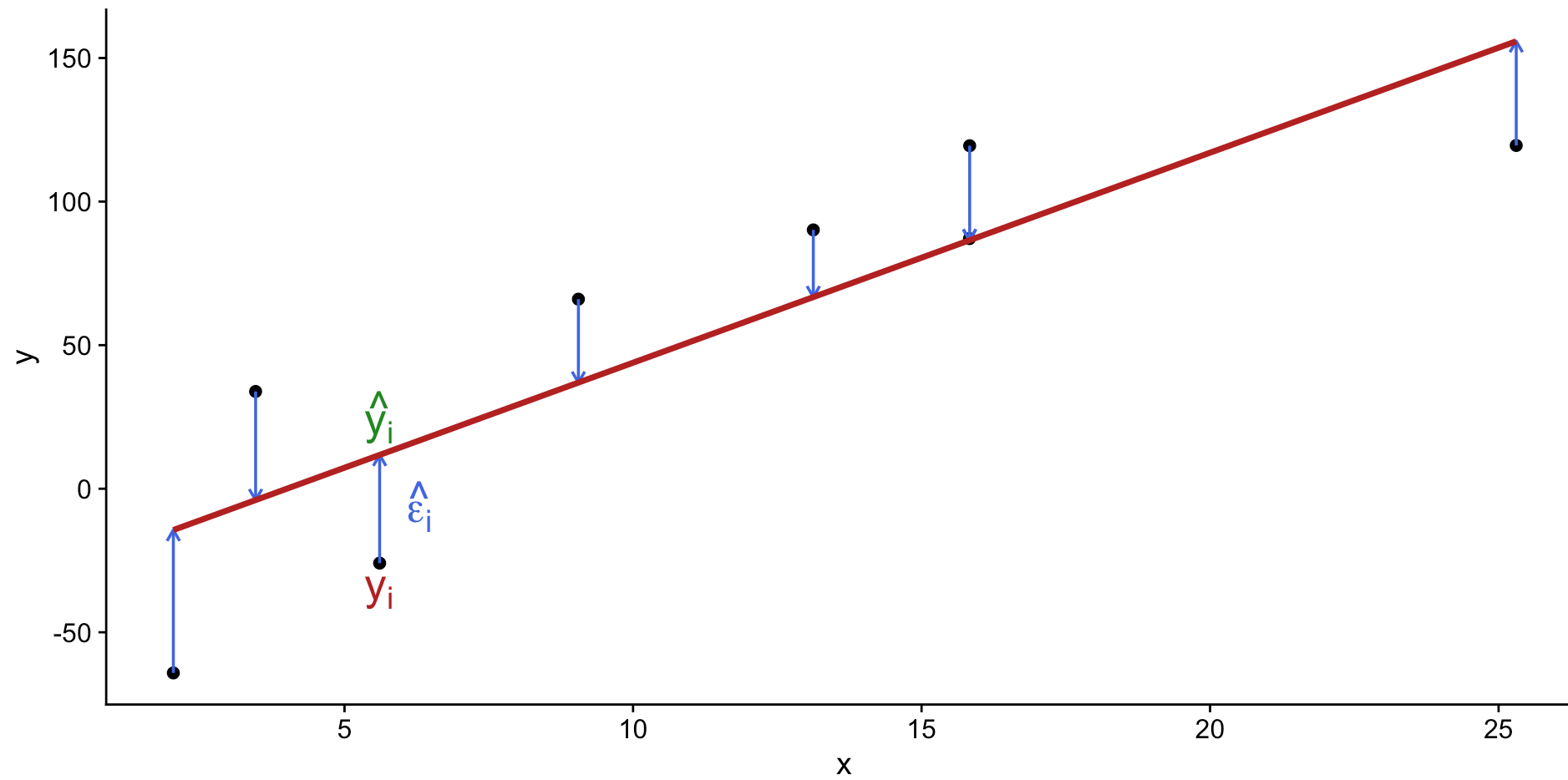
Least squares

The method of least squares is the **automobile of modern statistical analysis**: despite its limitations, occasional accidents and incidental pollution, it and its numerous variations, extensions, and related conveyances **carry the bulk of statistical analyses**, and are known and valued by nearly all.

– Stigler, 1981 (emphasis added)

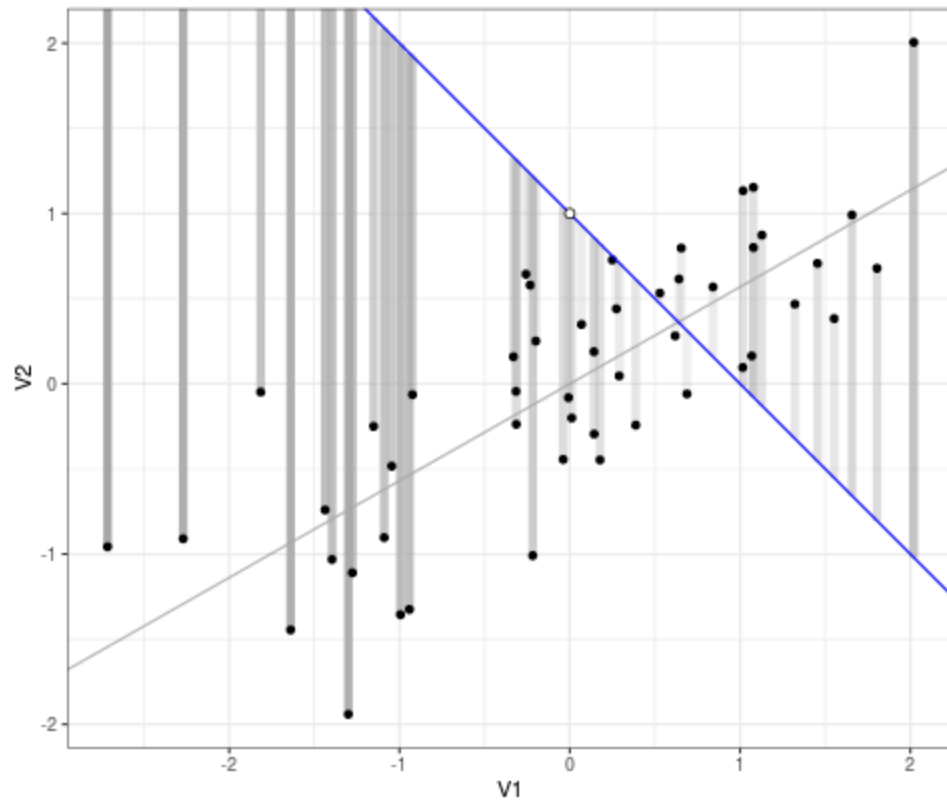
Residuals,

If a child is 178 cm tall and the fitted line predicts 174 cm, the residual is +4 cm.



How do we fit a line?

Choose the intercept () and slope () that make the total squared residual as small as possible:



1. Draw a candidate line.
2. Calculate the residuals.
3. Square and sum them.
4. Find the line with the smallest sum.
5. **Report its slope and intercept.**

Source: [ls-springs](#)

The linear model

Linear regression

A linear model describes the relationship between a response and a predictor .

- : intercept
- : slope
- : deviation of observation from the mean response predicted at

Modelling

In the model:

The blue component is the fixed mean relationship. The red component is random variation around it.

We can think of the response as being made up of two components:

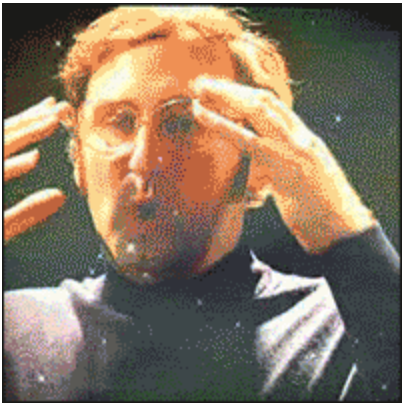
- Response = Prediction + Error
- Response = Signal + Noise
- Response = Deterministic + Random
- Response = Explainable + Everything else

How do we assess a fitted model?

A fitted line is not automatically a valid model.

We cannot observe the errors directly, so after fitting we inspect their estimates, the residuals .

Residual checks help us judge whether the observed error patterns are reasonably compatible with the model assumptions.



It is just about the errors... source: Tim & Eric Awesome Show, Great Job! Episode 3 of season 4.

From model to fitted equation

The model:

After fitting the data:

The fitted intercept is .

The fitted slope is .

For Galton's data:

Galton regression output

Term	Estimate (SE)	95% CI	p-value
Coefficients			
Intercept	23.94 (2.81)	18.43–29.46	<0.001
Parent height (in)	0.65 (0.041)	0.57–0.73	<0.001
Model fit			
R ²	0.21		
Adjusted R ²	0.21		
n	928		
SE = standard error; CI = confidence interval.			

Interpreting the fitted model

Main result: Parent height is positively associated with child height.

- **Slope:** For each 1-inch increase in parent height, predicted child height increases by 0.65 inches (95% CI 0.57–0.73,).
- **Model fit:** Parent height explains 21% of the variation in child height ().
- **Intercept:** 23.94 is the predicted child height when parent height is zero. Because zero is outside the observed range, it is not biologically meaningful here.

Example 2: Adelie penguins

► Code

Characteristic	Beta	95% CI	p-value
(Intercept)	-2,536	-4,442, -629	0.009
Flipper length (mm)	33	23, 43	<0.001
R ²	0.219		
Adjusted R ²	0.214		
Abbreviation: CI = Confidence Interval			

Among Adelie penguins, longer flippers were associated with greater body mass. For each 1 mm increase in flipper length, the model estimated a 33 g increase in body mass (95% CI: 23–43 g per mm,). The model accounted for 22% of the observed variation in body mass ().

Example 3: *Iris setosa*

► Code

Characteristic	Beta	95% CI	p-value
(Intercept)	1.2	0.73, 1.6	<0.001
Sepal width (cm)	0.08	-0.05, 0.21	0.2
R ²	0.032		
Adjusted R ²	0.011		
Abbreviation: CI = Confidence Interval			

In *Iris setosa*, the estimated association between sepal width and petal length was small and uncertain. A 1 cm increase in sepal width corresponded to an estimated 0.08 cm increase in petal length (95% CI: -0.05–0.21 cm per cm,). Because the interval includes zero, these data do not provide clear evidence of a linear association. The model accounted for only 3.2% of the observed variation ().

Example 4: Air quality data

► Code

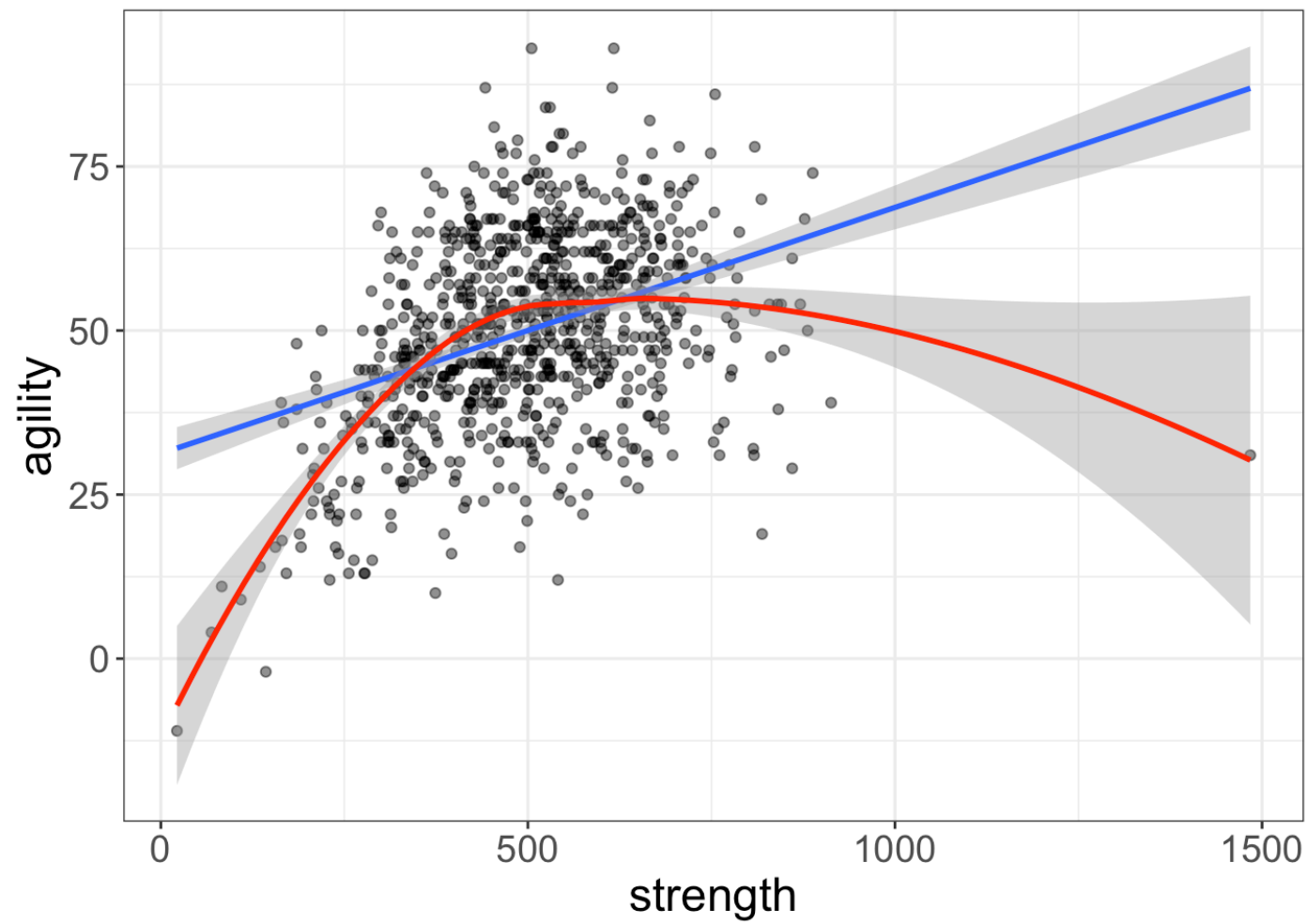
Characteristic	Beta	95% CI	p-value
(Intercept)	10	9.1, 12	<0.001
Solar radiation (Langley units)	0.00	-0.01, 0.00	0.5
R ²	0.003		
Adjusted R ²	-0.004		
Abbreviation: CI = Confidence Interval			

In the 1973 New York *airquality* data, solar radiation was not a useful linear predictor of wind speed. The estimated slope was -0.002 mph per Langley (95% CI: -0.009 – -0.004 mph per Langley,), and the model accounted for only 0.3% of the observed variation (). This does not prove that the variables can never be related.

A fitted model is not automatically a good model

Any dataset can produce a fitted line. Before interpreting its coefficients, ask:

- Does the line capture the systematic pattern in the data?
- Do the residuals behave as expected?



Comparison of two candidate models fitted to the same data, shown in red and blue. The better model captures the pattern with less systematic residual structure.

Quick check: Which model better represents the pattern: red or blue?

Next week: model assumptions

We will explore the assumptions of linear regression and learn how to assess whether they are reasonable for our data.

Thanks

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